Exercise - 2.4

1. Construct a quadratic equation with roots 7 and -3.
2. A quadratic polynomial has one of its zeros $1+\sqrt{5}$ and it satisfies $p(1)=2$. Find the quadratic polynomial.
3. If α and β are the roots of the quadratic equation $x^2+\sqrt{2}x+3=0$, form a quadratic polynomial with zeroes $\frac{1}{\alpha}, \frac{1}{\beta}$.
4. If one root of $k(x-1)^2=5x-7$ is double the other root, show that $k=2$ or -25.
5. If the difference of the roots of the equation $2x^2-(a+1)x+a-1=0$ is equal to their product, then prove that $a=2$.
6. Find the condition that one of the roots of ax^2+bx+c may be (i) negative of the other, (ii) thrice the other, (iii) reciprocal of the other.
7. If the equations $x^2-ax+b=0$ and $x^2-ex+f=0$ have one root in common and if the second equation has equal roots, then prove that $ae=2(b+f)$.
8. Discuss the nature of roots of (i) $-x^2+3x+1=0$, (ii) $4x^2-x-2=0$, (iii) $9x^2+5x=0$.
9. Without sketching the graphs, find whether the graphs of the following functions will intersect the x-axis and if so in how many points.
   (i) $y=x^2+x+2$, (ii) $y=x^2-3x-7$, (iii) $y=x^2+6x+9$.
10. Write $f(x)=x^2+5x+4$ in completed square form.

2.5.2 Quadratic Inequalities

Here we shall learn to solve the quadratic inequalities $ax^2+bx+c<0$ or $ax^2+bx+c>0$.

Steps to Solve Quadratic Inequalities:

(i) First solve $ax^2+bx+c=0$.
(ii) If there are no real solutions, then one of the above inequality holds for all $x\in\mathbb{R}$
(iii) If there are real solutions, which are called *critical points*, then label those points on the number line.
(iv) Note that these critical points divide the number line into disjoint intervals. (It is possible that there may be only one critical point.)
(v) Choose one representative number from each interval.
(vi) Substitute that these representative numbers in the inequality.
(vii) Identify the intervals where the inequality is satisfied.

Example 2.13 Solve $3x^2+5x-2\leq 0$.

Solution:
On factorizing the quadratic polynomial we get $3(x+2)(x-\frac{1}{3})\leq 0$. Draw the number line. Mark the critical points -2 and $\frac{1}{3}$ where the factors vanish (See Figure 2.5). On each sub-interval check the sign of $(x+2)(x-\frac{1}{3})$. To do this pick an arbitrary point anywhere in the interval. Whatever sign the resulting value has, the polynomial has the same sign throughout the whole corresponding interval. (Otherwise, there would be another critical point within the interval). This process is easily organized in the following table.